7.2 Polarizers

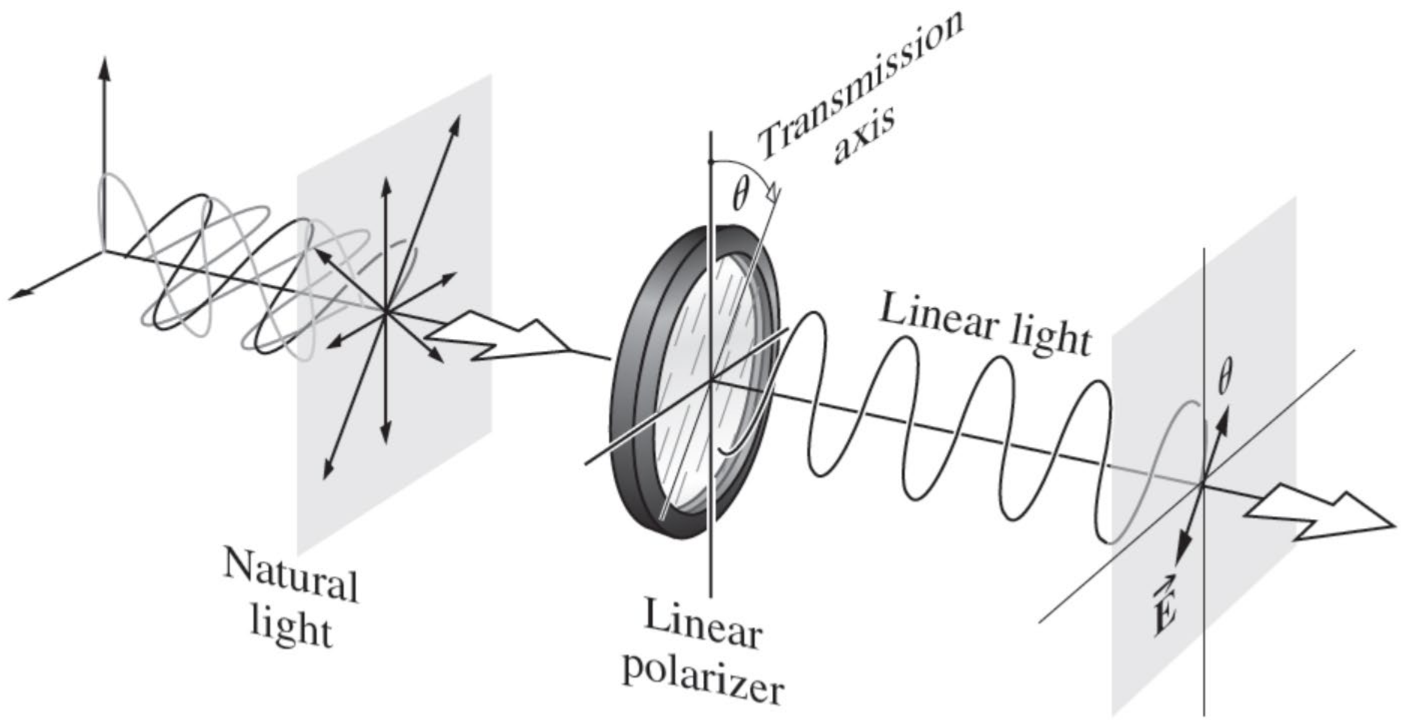
Polarizer: an optical device whose input is natural light and whose output is some form of polarized light.

<u>Linear polarizer; circular polarizer; elliptical polarizer and partial polarizer.</u>

Fundamental physical mechanisms of polarizer:

- ❑ Dichroism (二向色性)/ selective absorption;
- ❑ Reflection;
- ❑ Scattering;
- ❑ Birefringence (双折射)/ double refraction

Optical transmission of a linear polarizer

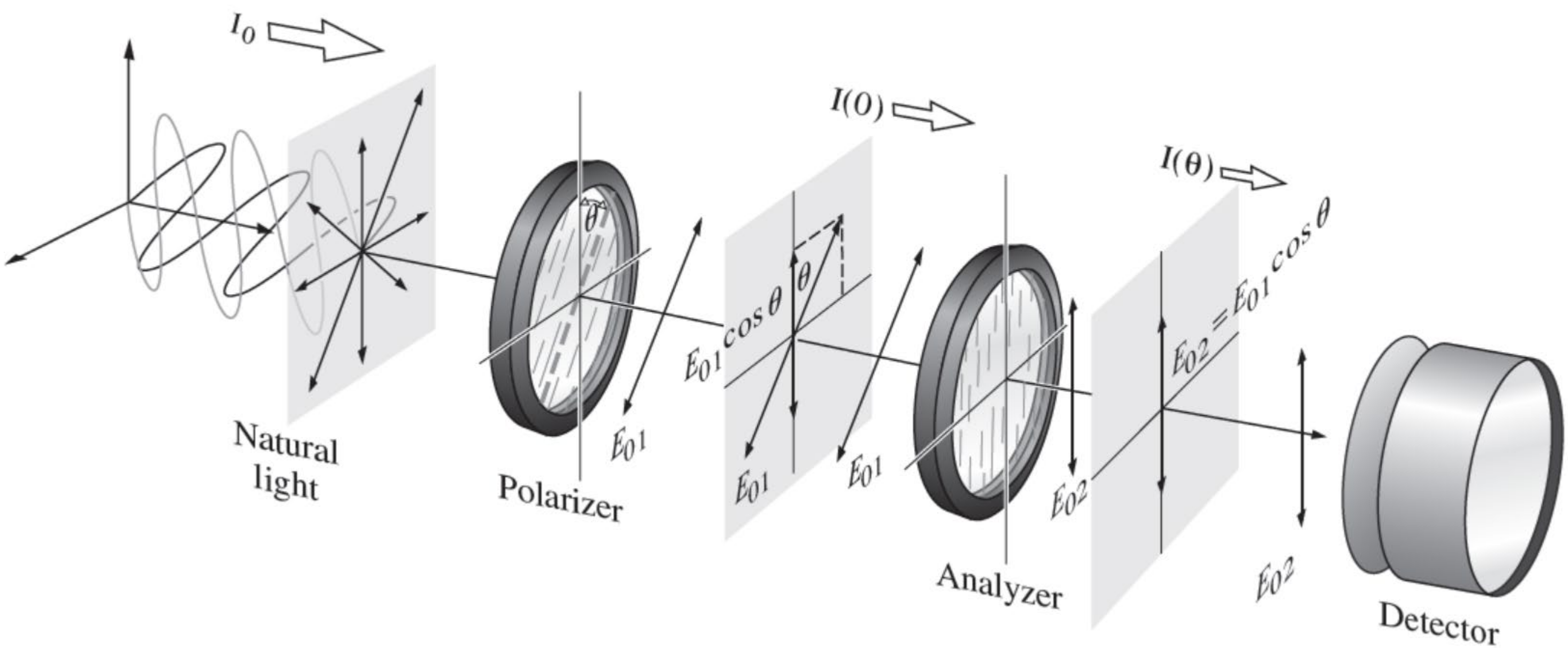


We can only measure intensity (irradiance) $I = \frac{c\epsilon_0}{2}E^2$

Malus's law (马吕斯定律)

Intensity of transmitted light through a linear polarizer

$$I(\theta) = \frac{c\epsilon_0}{2} {E_{01}}^2 \cos^2\theta = I(0)\cos^2\theta$$

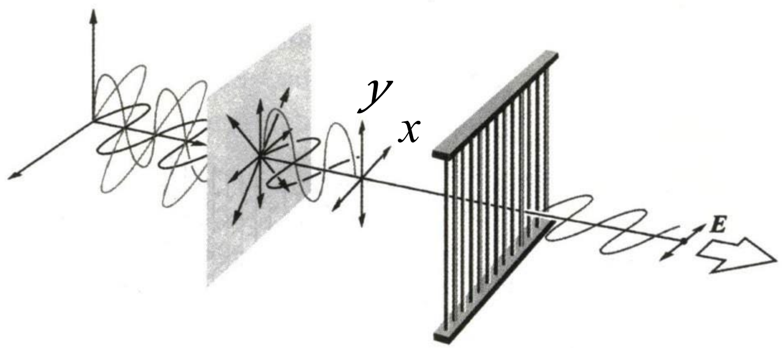


7.3 Dichroism

Dichroism: the selective absorption of one of the two orthogonal $\mathcal{P}$-state components of an incident beam.

The wire-grid polarizer

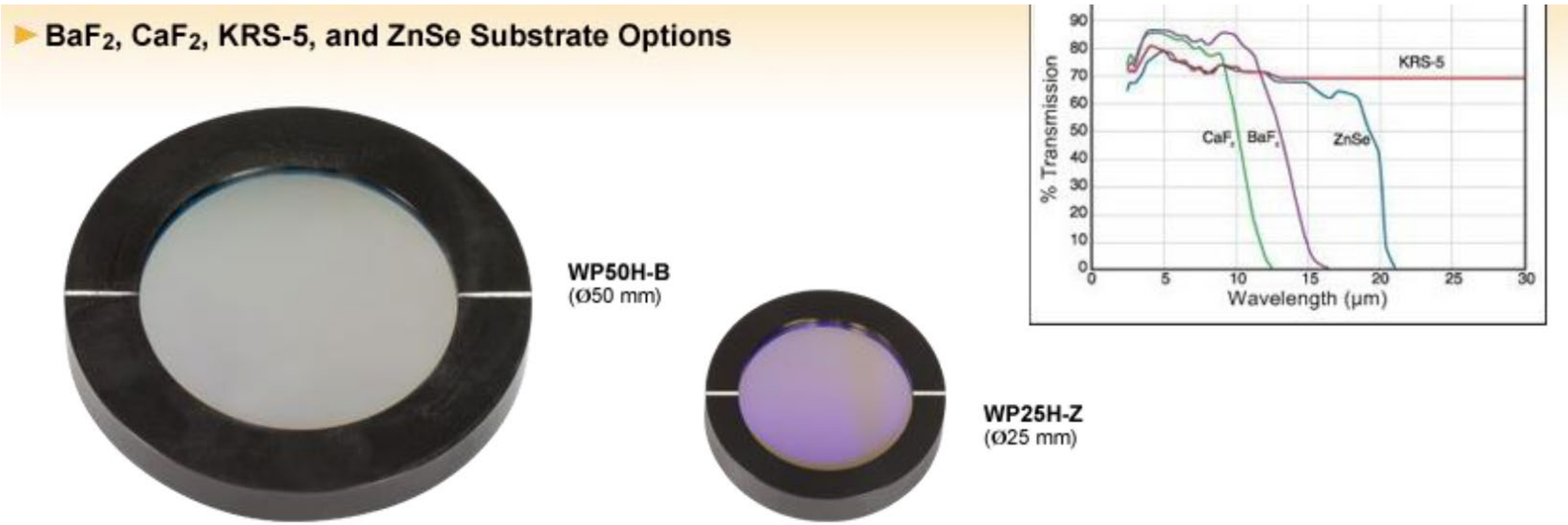
The simplest device is a grid of parallel conducting wires.



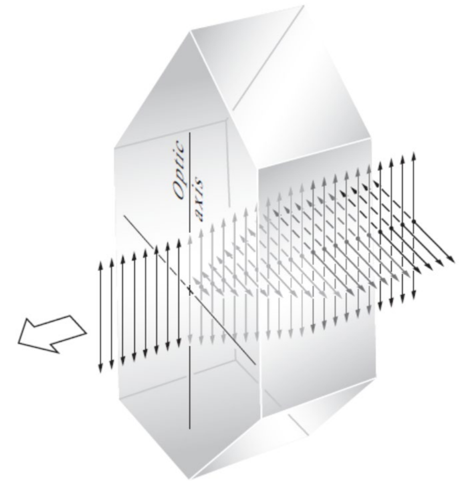
How it works:

- ❑ The y-component of the field → drive conduction electrons along the length of each wire → collide with lattice atoms (joule heat) → energy is transferred from the field to grid.
- ❑ Electrons accelerating along the y-axis → radiate in the forward and backward direction → the incident wave cancel the wave reradiated in the forward direction → no transmission of the y-component of field
- ❑ The electrons are not free to move very far in x-component → no absorption
- ❑ The transmission axis of the grid is perpendicular to the wires

Commercial product



Dichroic Crystals Certain materials are inherently dichroic because of an anisotropy in their respective crystalline structure.



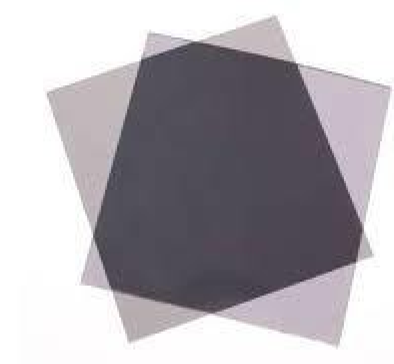
For example: tourmaline (电气石)

- ❑ Principal or optical axis determined by its atomic configuration: electric-field component perpendicular to the principal axis is strongly absorbed.
- ❑ The crystal’s principal axis becomes the polarizer’s transmission axis.

Polaroid (人造偏振片):

<u>J-sheet</u>: very small needle-shaped herapathite (碘硫酸奎宁) were aligned nearly parallel by means of magnetic or electric fields or narrow slits.

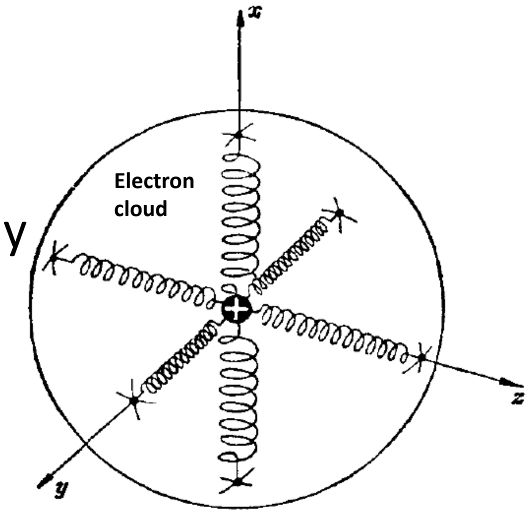
<u>H-sheet</u>: it contains aligned molecules [polyvinyl alcohol (聚乙烯乙醇) with iodine] analogue of the wire grid.



Homework

- 8.1, 8.3, 8.21

7.4 Birefringence

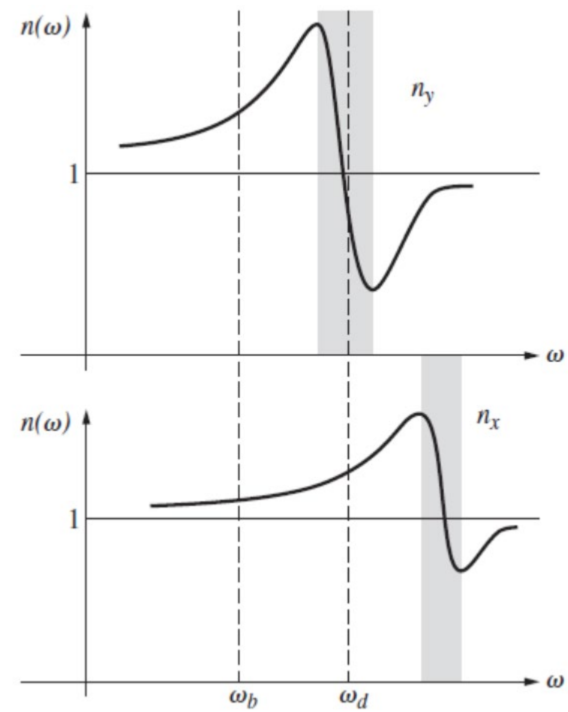


Many crystalline substances are optically anisotropic. It can be described using a mechanical model depicting a negatively charged shell bound to a positive nucleus by pairs of springs having different stiffness.
Model: The refractive index *n*

$$n^2(\omega) = 1 + \frac{Nq^2}{\epsilon_0 m}\left(\frac{1}{{\omega_0}^2 - \omega^2}\right)$$

ω and ω_0: the angular frequency of the driving force and the natural angular frequency of the electric dipole, respectively.

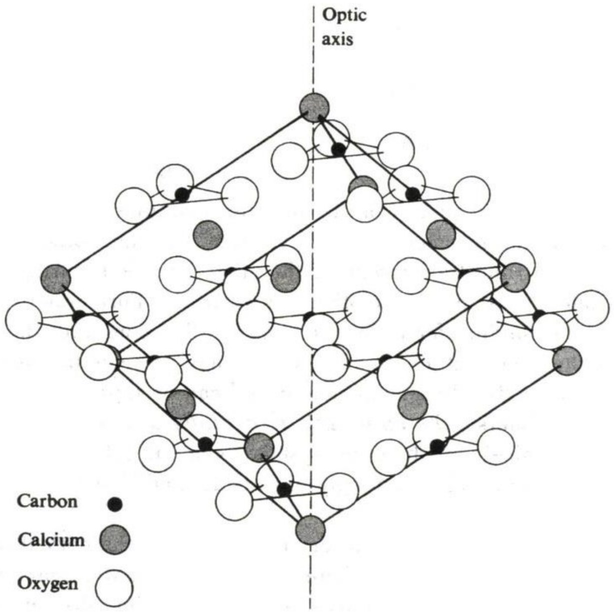
In the vicinity of ω_d, one of the orthogonal $\mathcal{P}$-state is absorbed and the other passes. It is **dichroic**.

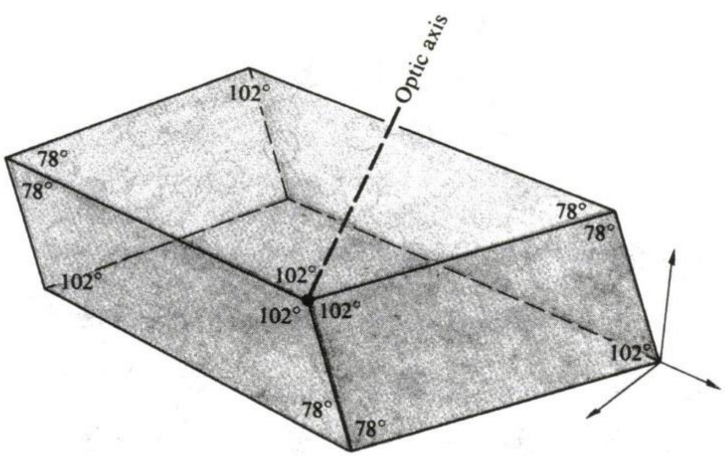


If the stiffness of the spring in x direction has one value while has another value along any direction perpendicular to x-direction, the material has two different refractive indices (In the vicinity of ω_d) and is said to be **birefringent**. The x-direction is called the **optic axis**.

Calcite（方解石）

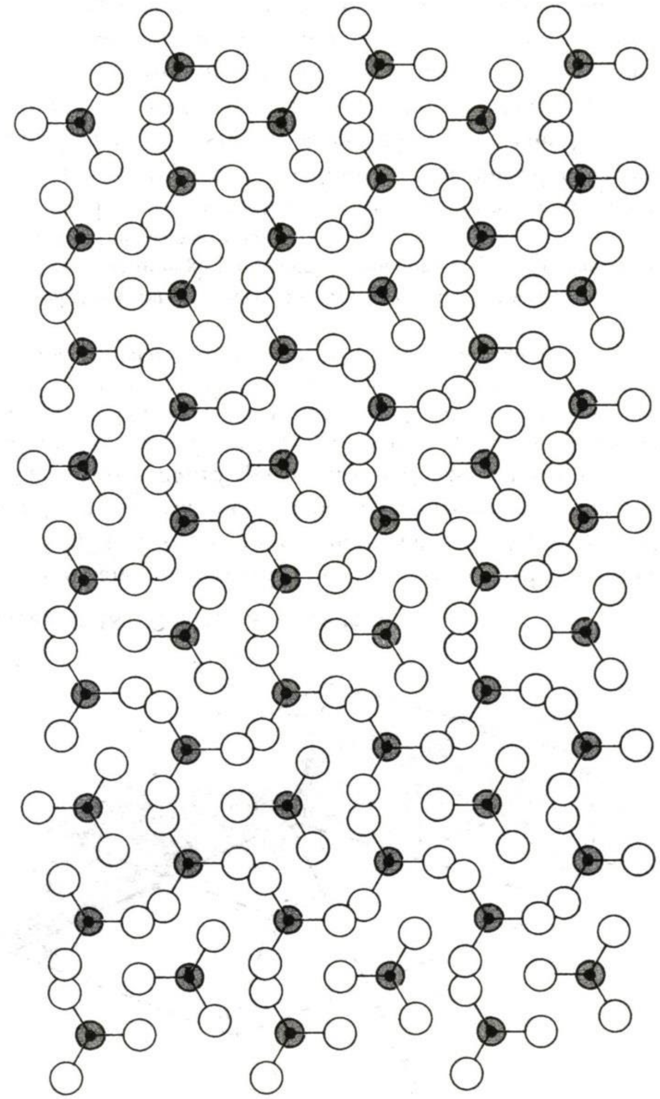
A typical birefringent crystal （$CaCO_3$）



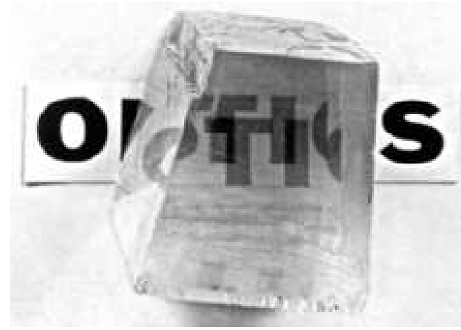


Cleavage form

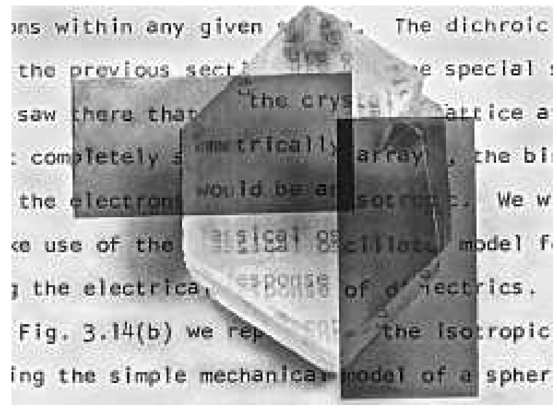
Rhombohedron (斜方六面体)



Ordinary rays (o-rays)
Extraordinary rays (e-rays)

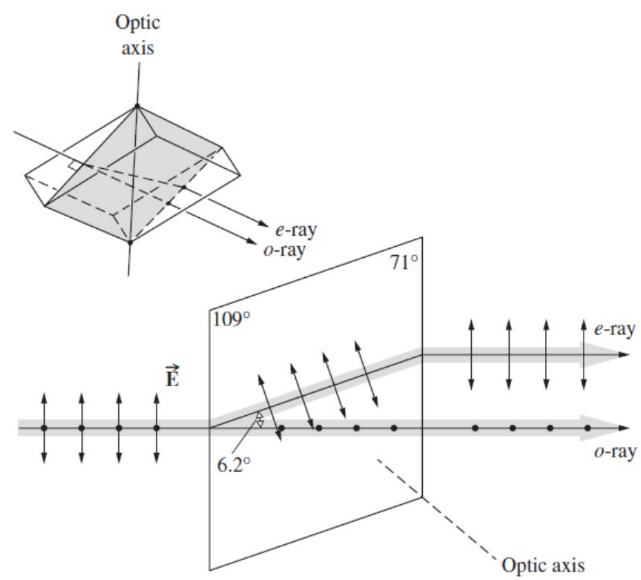


The ordinary and extraordinary images are linearly polarized, and two $\mathcal{P}$-state are orthogonal.

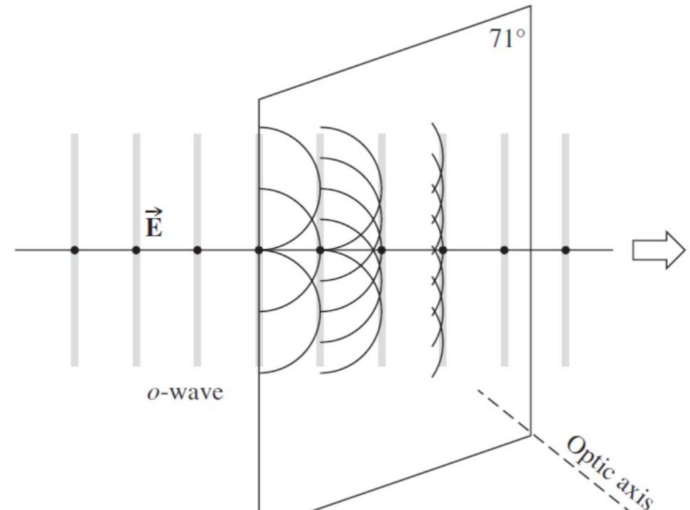


Principal planes (主平面): any number of planes is drawn through the rhomb so as to contain the optic axis.

Principal section (主截面): if the principal plane is normal to a pair of opposite surfaces of the cleavage form.

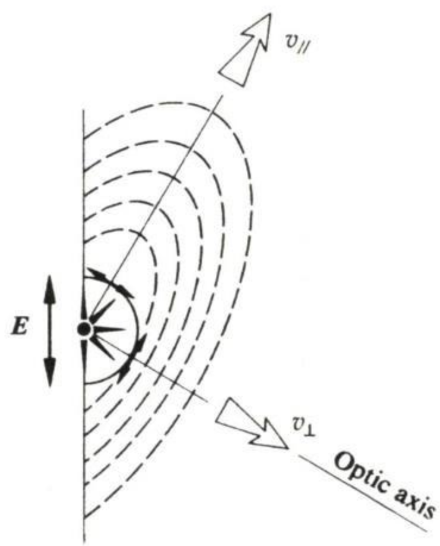


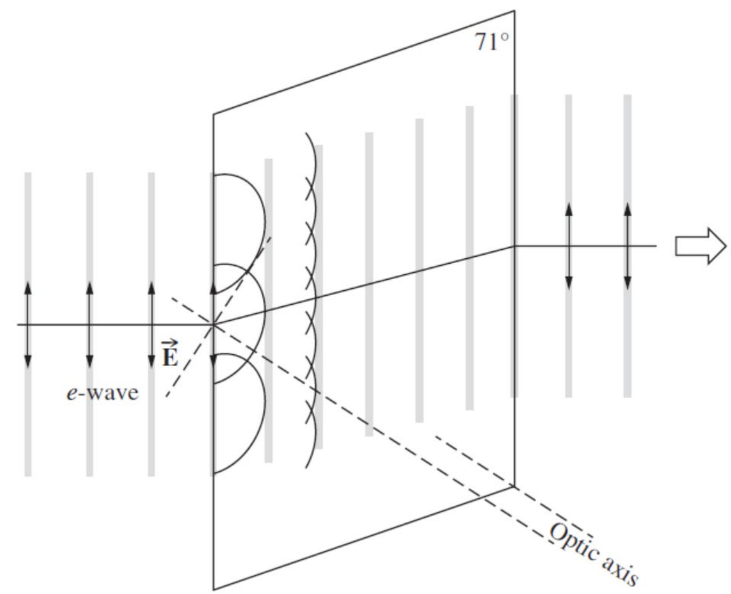
The incident plane wave is linearly polarized and perpendicular to the optic axis.
The velocity $v_\perp$ in all direction is the same.



E-field is parallel to the principal section:

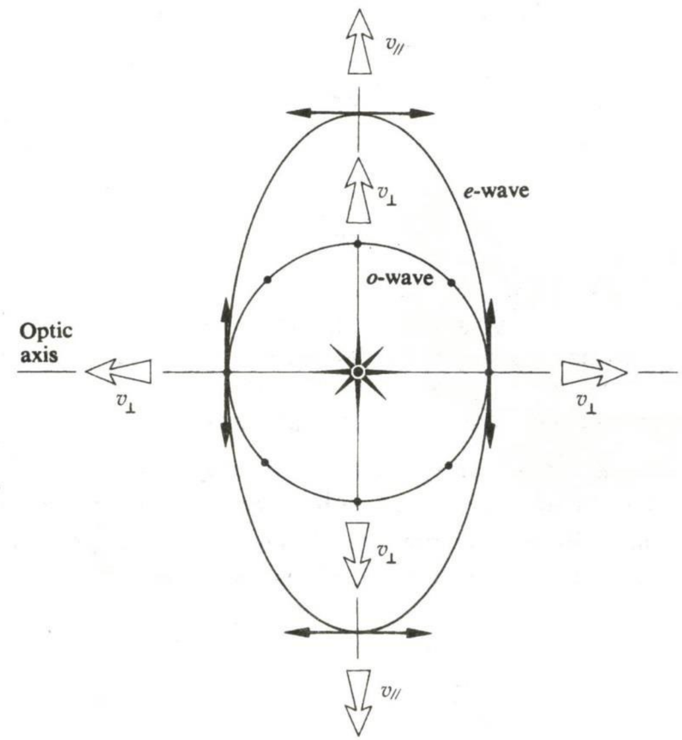
- ❑ a component normal to the optic axis $v_\perp$;
- ❑ a component parallel to it $v_\parallel$ But $v_\perp \neq v_\parallel$





Birefringent crystals

The optic axis: a direction about which the atoms are arranged symmetrically.
Uniaxial: crystals with only one such direction.



Ordinary rays (o-rays): The E-field of the o-wave is everywhere normal to the optic axis, so it moves at a speed $v_\perp$ in all directions.

Extraordinary rays (e-rays):

- The E-field of e-wave has a speed $v_\perp$ only in the direction of the optic axis
- E-field parallel to the optic axis: speed $v_\parallel$

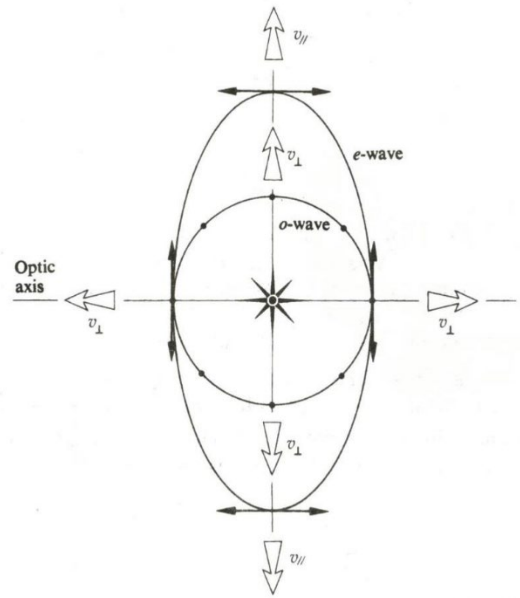
Uniaxial materials have two principle indices of refraction:

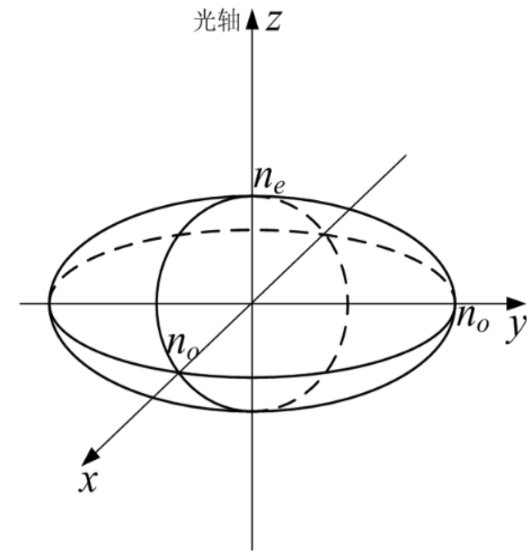
$$n_0 \equiv \frac{c}{v_\perp} \qquad n_e \equiv \frac{c}{v_\parallel}$$

Birefringence: the difference $\Delta n = n_e - n_o$

Calcite $v_{\parallel} > v_{\perp}$, $\Delta n = -0.172$, negative uniaxial.

Calcite $n_o = 1.658$; $n_e = 1.486$

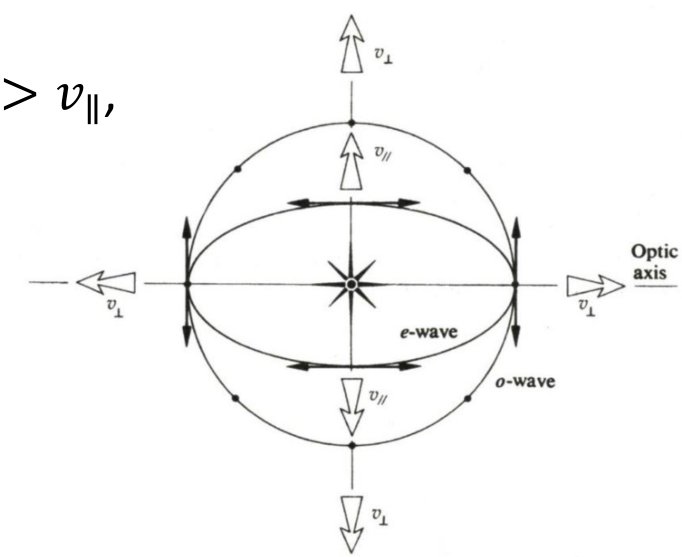


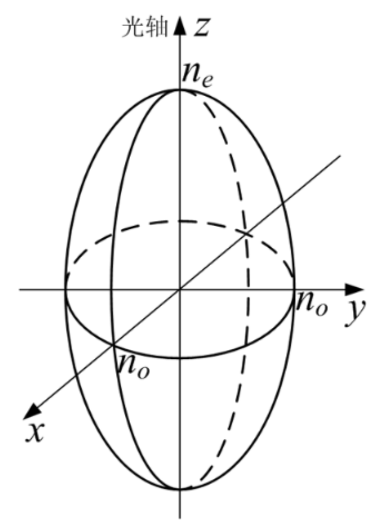


负单轴晶体

Other crystals (quartz and ice):$v_{\perp} > v_{\parallel}$, Δn is positive, positive uniaxial.

Quartz $n_o = 1.544$; $n_e = 1.553$





正单轴晶体

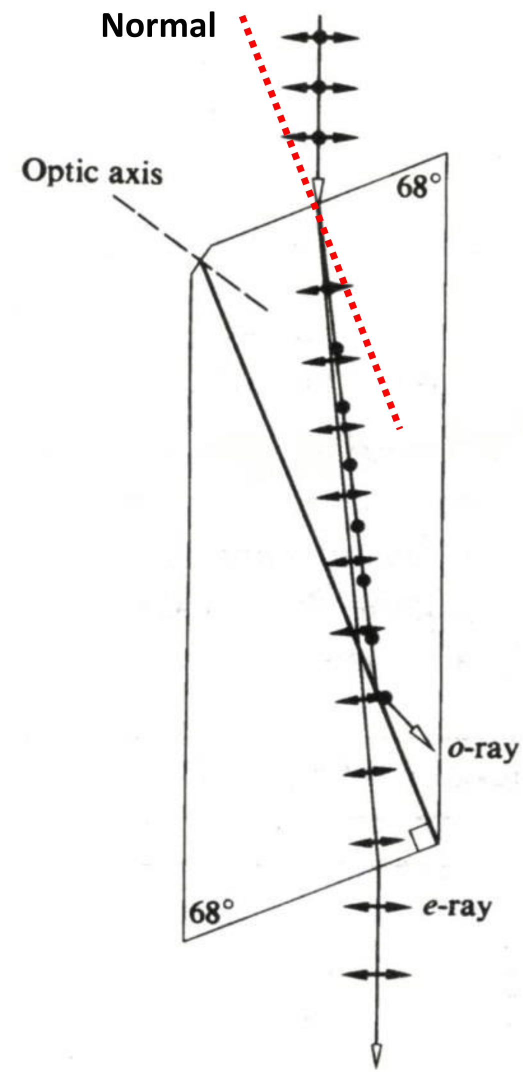
Birefringent polarizers

Nicol prism:
The device is made by first grinding and polishing the ends of a suitable long, narrow calcite rhombohedron.

Calcite $n_o = 1.658;\ n_e = 1.486; n_o > n_e;\ \theta_{to} < \theta_{te}$

The two pieces are polished and cemented back together with Canada balsam (加拿大树胶) with an index of 1.55, which almost in the midway between n_e and n_o.

The incident beam are refracted and separated after entering the prism. On the interface, o-ray is totally internally reflected and the e-ray passes.

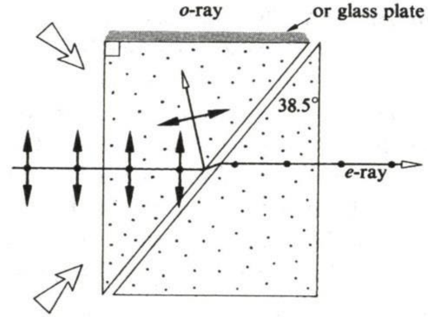


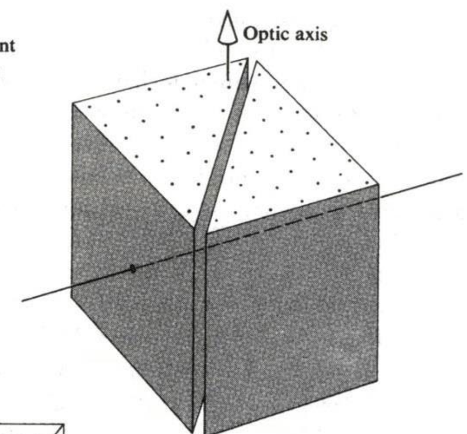
Glan-Foucault polarizer (Glan-Air polarizer): calcite (5000nm-230nm)

The incoming ray strikes the surface normally, including two components (parallel and perpendicular to the optic axis)

In the first section: two rays traverse without any deviation.

In the calcite-air interference: set $n_e < \frac{1}{\sin\theta} < n_o$, o-ray is totally internally reflected; e-ray passes.

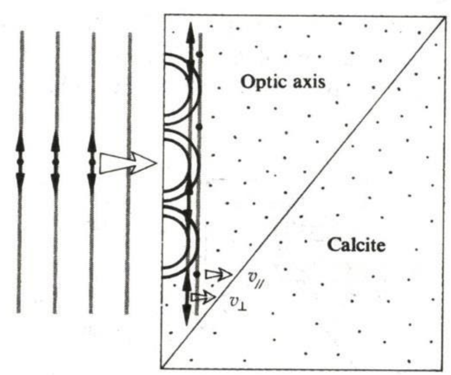




Glan-Thompson polarizer:

Two prisms are cemented together and interface angle is changed appropriately.

- ❑ Advantage: High field of view (30°), for Glan-Foucault polarizer, 10°.
- ❑ Disadvantage: can not handle high power levels of lasers

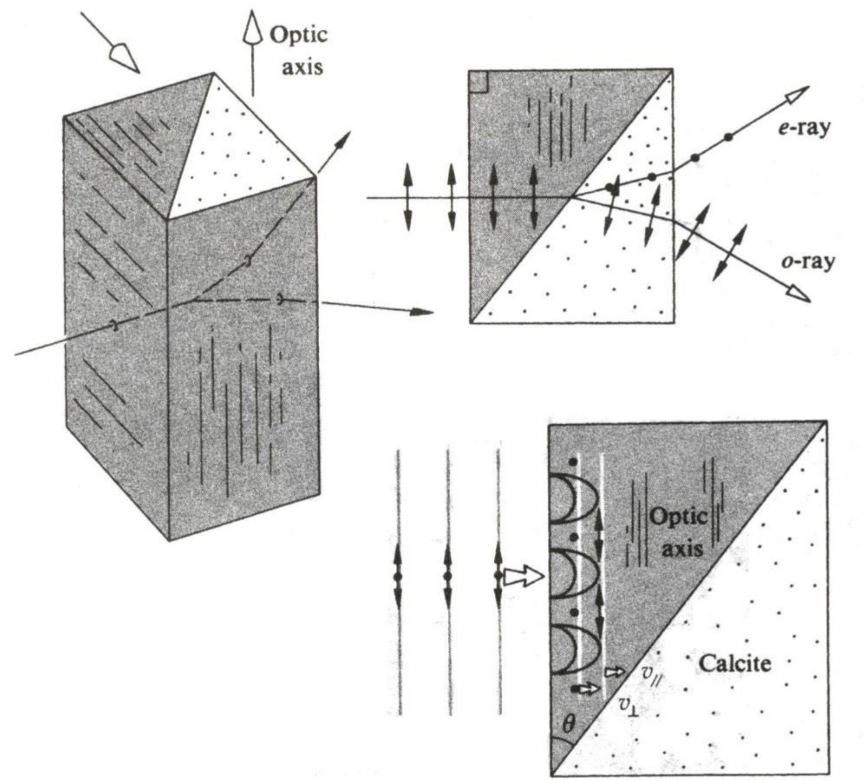


Wollaston prism:

A polarizing beamsplitter, (calcite or quartz).

In calcite: $n_e < n_o$

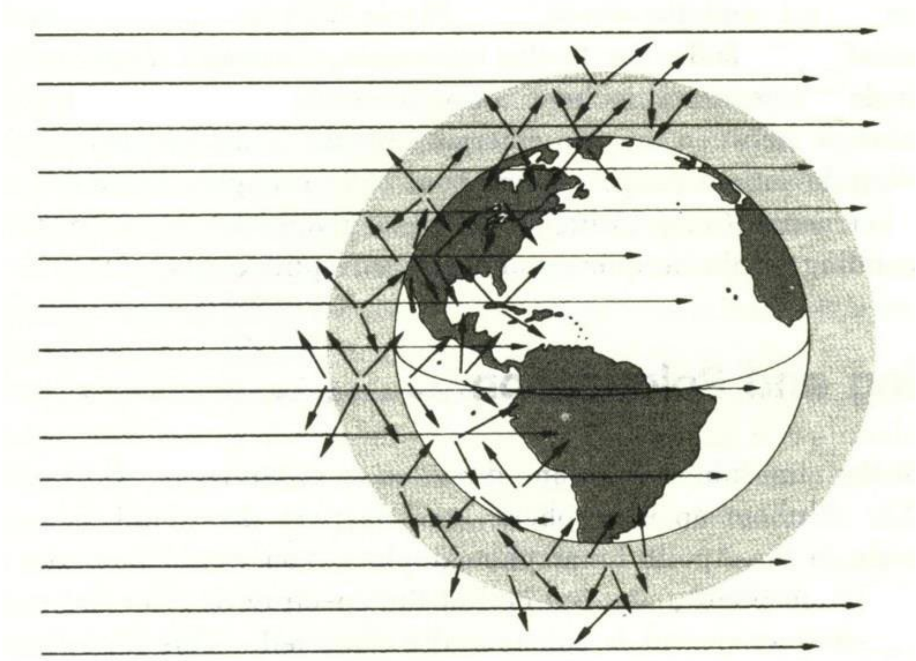
The o-ray, whose field is initially perpendicular to the optic axis, becomes an e-ray, which is bent away from the normal to the interface.



Homework

- 8.40

7.5 Scattering and Polarization

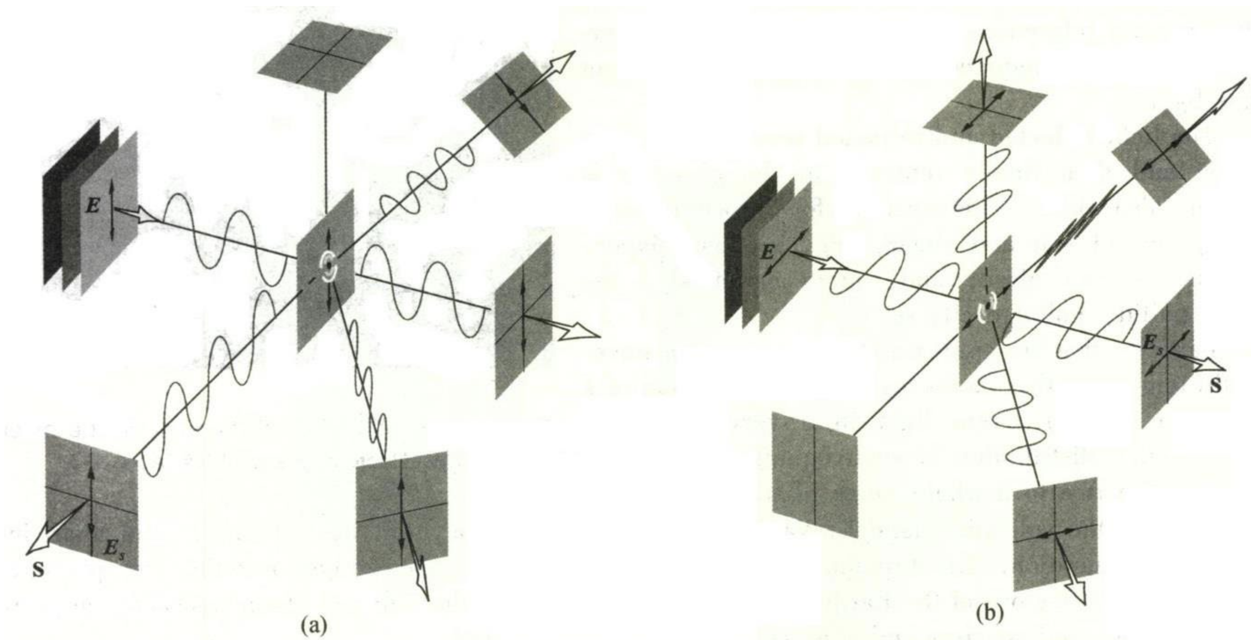


Sunlight streaming into the atmosphere from one direction is scattered in all directions by the air molecules.

7.5.1 Polarization by Scattering

A linearly polarized plane wave incidents on an air molecule;

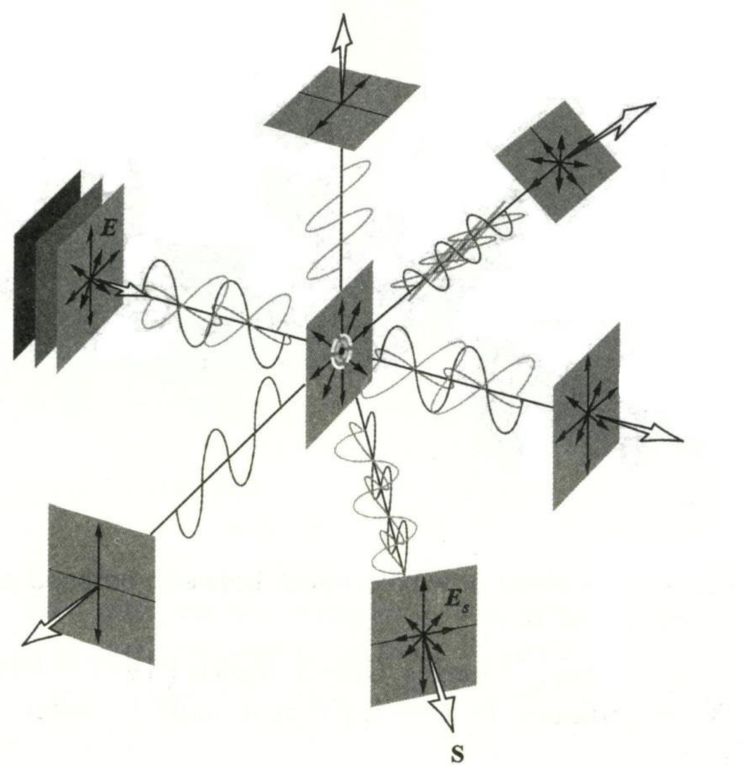
- The vibrations induced in the atom are parallel to the E-field of the incoming lightwave and perpendicular to the propagation direction.
- There is no radiation in the direction of its axis.



Scattering of polarized light by a molecule.

As to unpolarized light:

- The scattered light in the forward direction is completely unpolarized;
- Off that axis it is partially polarized;
- When the direction of observation is normal to the primary beam, the light is completely linearly polarized.



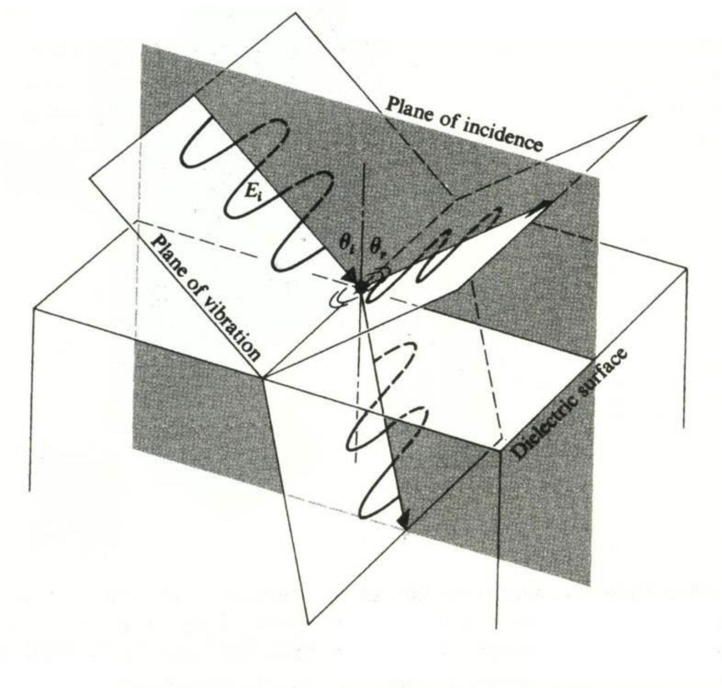
Scattering of unpolarized light by a molecule.

7.6 Polarization by Reflection

The Electron Oscillator Model:

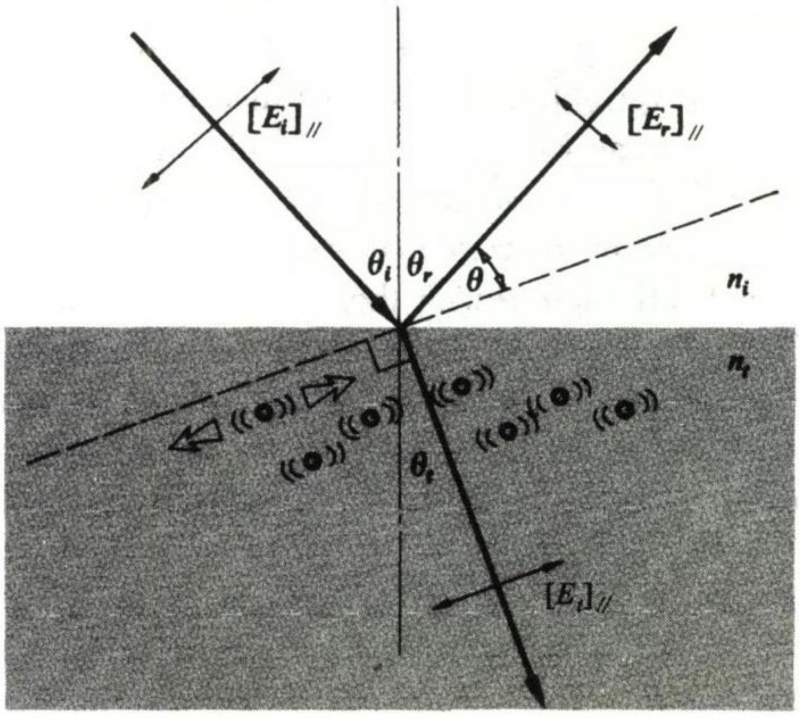
<u>The E-field is perpendicular to the plane of incidence</u>

- The wave is refracted at the interface, and enter the medium at some transmission angle.
- Its electric field drives the bound electrons, and they in turn reradiate.
- A portion of that reemitted energy appears in the form of a reflected wave. And both the reflected and refracted waves must also be in P-states normal to the incident plane.



The E-field is in the incident plane

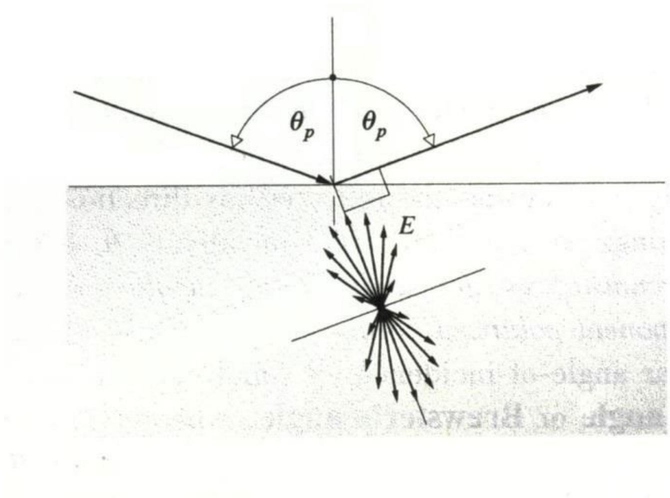
- The electron-oscillators near the surface will vibrate under the influence of the refracted wave.
- The reflected ray direction makes a small angle with the dipole axis and the flux density of reflected wave is relatively low.
- When θ=0, the reflected wave would vanish entirely.

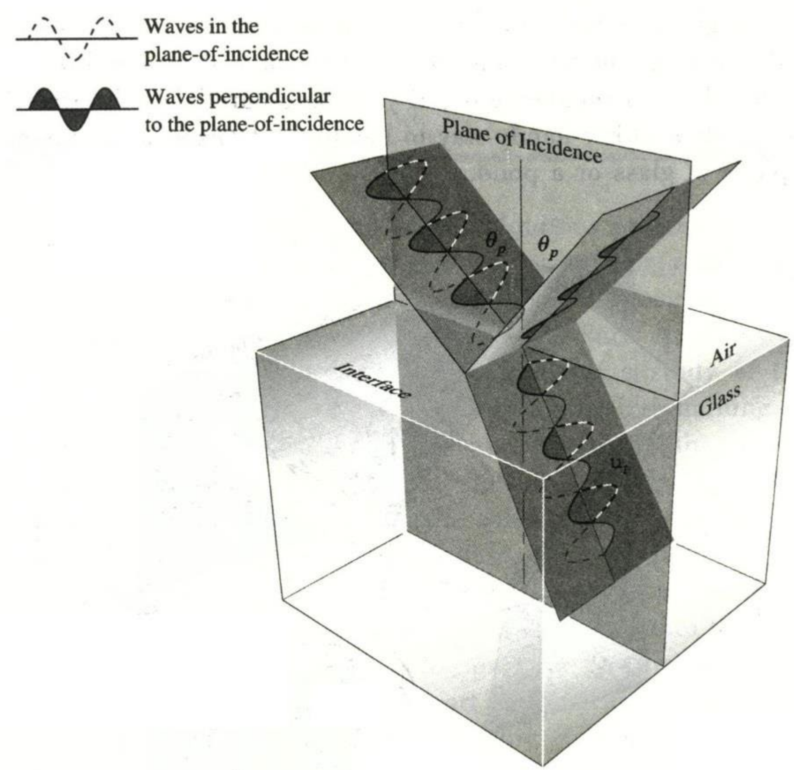


An incoming unpolarized wave(two incoherent orthogonal P-state): only the component polarized normal to the incident plane will be reflected for the **polarization angle or Brewster's angle**.

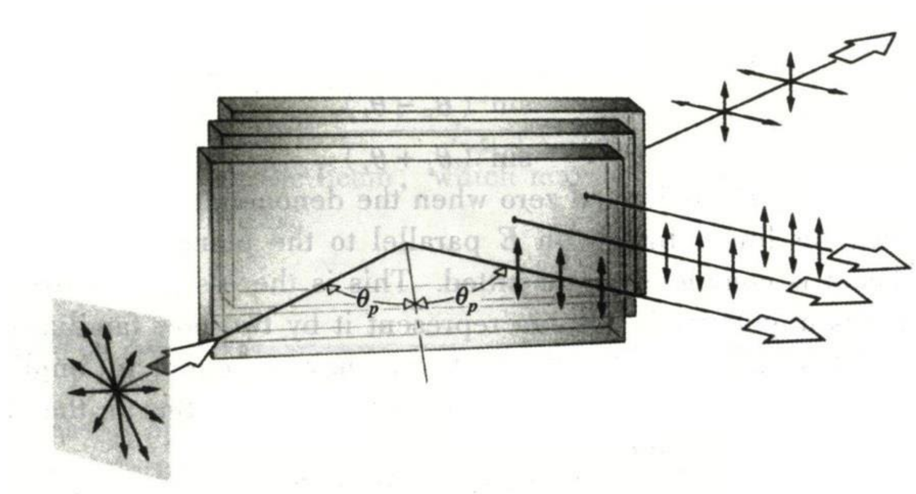
When $\theta_p + \theta_t = 90°$,
From Snell's Law: $n_i \sin\theta_p = n_t \sin\theta_t$
We have: $\tan\theta_p = n_t/n_i$

Example:
For air/glass ($n_t \approx 1.5$), the polarization angle is 56°,
For air/pond ($n_t \approx 1.33$), the polarization angle is 53°.

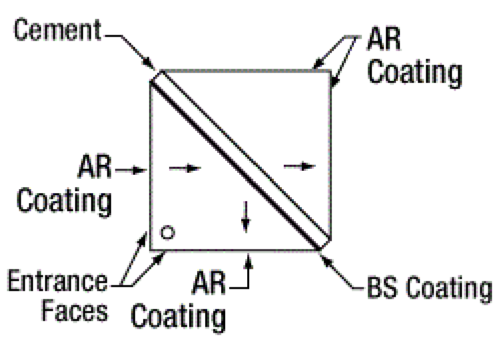


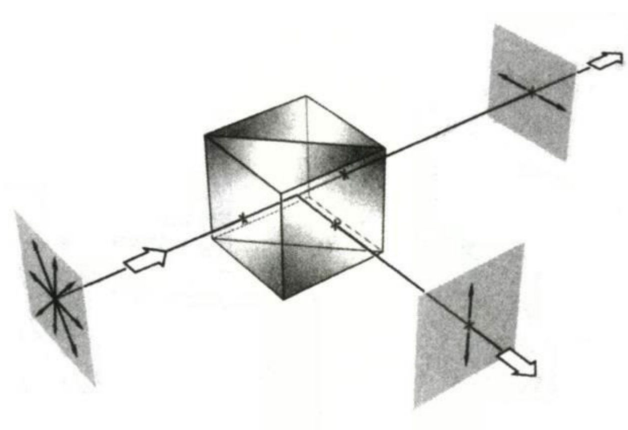


The pile-of-plates polarizer



A polarizing cube contains a mutilayer dielectric thin film structure on its diagonal face.





PBS253
(1")

7.6.1 An Application of the Fresnel Equations

For linear light having its E-field parallel to the plane-of-incidence:

Amplitude reflection coefficient: $r_{\parallel} \equiv [E_{0r}/E_{0i}]_{\parallel}$

For E-field normal to the incident plane:

Amplitude reflection coefficient: $r_{\perp} \equiv [E_{0r}/E_{0i}]_{\perp}$

Reflectance:$R_{\parallel} = {r_{\parallel}}^2 = \ {[E_{0r}/E_{0i}]_{\parallel}}^2$ and $R_{\perp} = {r_{\perp}}^2 = \ {[E_{0r}/E_{0i}]_{\perp}}^2$

Squaring the appropriate Fresnel Equations yields:

$$R_{\parallel} = \frac{\tan^2(\theta_i - \theta_t)}{\tan^2(\theta_i + \theta_t)}$$

$$R_{\perp} = \frac{\sin^2(\theta_i - \theta_t)}{\sin^2(\theta_i + \theta_t)}$$

When $\theta_i + \theta_t = 90°$ (Brewster's angle), namely the denominator is infinite, $R_{\parallel}$ is indeed zero, $R_{\perp}$ can never be zero.

The incoming light is unpolarized ($I_{i\parallel} = I_{i\perp} = I_i/2$)

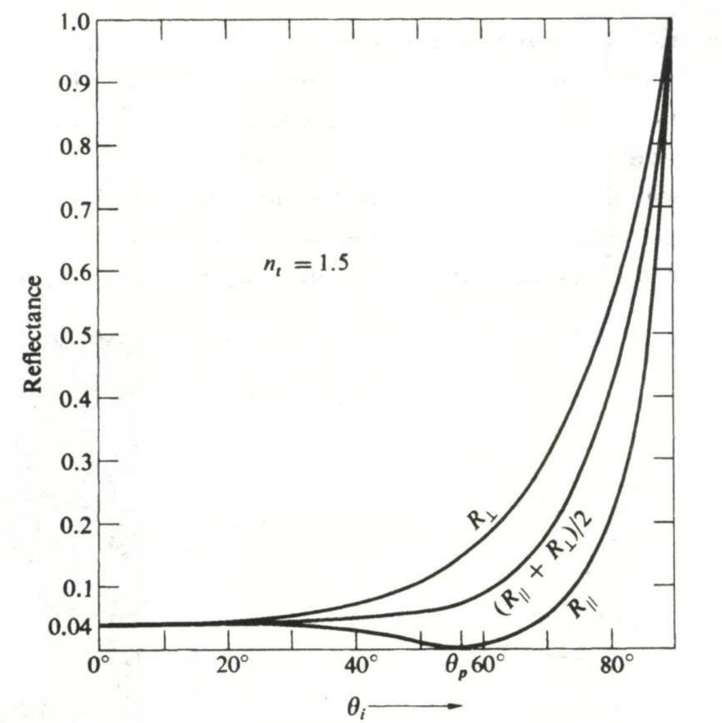
It is represented by two orthogonal, incoherent, equal-amplitude $\mathcal{P}$-state.

$$I_{r\parallel} = \frac{I_{r\parallel} I_i}{2 I_{i\parallel}} = R_\parallel I_i/2$$

$$I_{r\perp} = \frac{I_{r\perp} I_i}{2 I_{i\perp}} = R_\perp I_i/2$$

The reflectance in natural light:

$$R = \frac{I_r}{I_i} = \frac{I_{r\parallel} + I_{r\perp}}{I_i} = \frac{1}{2}\left(R_\parallel + R_\perp\right)$$



Note:

- ❑ When $\theta_i = \theta_p$, 7.5% of the incoming light is reflected;
- ❑ When $\theta_i \neq \theta_p$, both the transmitted and reflected waves are partially polarized.

Degree of polarization (偏振度) V: $$V = \frac{I_p}{I_p + I_u}$$

When using a linear polarizer to check a partially polarized light $$V = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$$